

Erdaifu (Carl) Luo

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EDUCATION

University of California, Berkeley

May 2028

BS, Applied Mathematics & BA, Computer Science

GPA: 3.8

- **Relevant Coursework:** Abstract Linear Algebra, Discrete Math & Probability, Abstract Algebra, Data Structures & Algorithms, Probability & Random Processes, Optimization Models, Computer Networks
- Awards: Deans List. Scholarships: The Leadership Award Scholar, Amazon Future Engineer

Diamond Bar High School

2020 – 2024

AIME Qualifier (2×); AMC 12 top 1% nationally; Highest scorer at HS (2023, 2024)

GPA: 4.0

EXPERIENCE

Amazon – Goodreads

June 2026 – Aug. 2026

Software Development Engineer Intern

Seattle, WA

- Building account recovery features within Goodreads' 6-million-line Ruby on Rails monolith, one of the largest Rails codebases in production.

Amazon

May 2025 – Aug. 2025

Software Development Engineer Intern

Seattle, WA

- Led development of an end-to-end automation solution for onboarding internal data services, cutting setup time from 40+ developer hours to minutes and eliminating human error.
- Designed and implemented custom API integrations and error-handling layers in Java to connect multiple internal platforms.
- Built secure cross-account access using AWS role assumption, enabling deployments across multiple regions and environments.
- Developed Python scripts for complex schema transformations to ensure compatibility across data pipelines.

Open Computing Facility, UC Berkeley

Aug. 2024 – Present

SysAdmin DeCal Head

Berkeley, CA

- Head and lead instructor for the SysAdmin DeCal since January 2025, a student-run course on Linux system administration and infrastructure, receiving 100+ applications and serving 40+ students per semester.
- Manage course logistics, oversee admissions, and maintain the DeCal website hosted on OCF servers, alongside other open-source software.
- Develop and deliver curriculum on server administration, networking, security, automation, and containerization, preparing students for professional sysadmin and DevOps roles.
- Collaborate with other OCF staff to operate and improve Berkeley's enterprise-scale computing infrastructure.

PROJECTS

Self-Hosted Homelab | *Docker, Kubernetes, Proxmox, Traefik, Authentik*

May 2024 – Present

- Designed and deployed a virtualized infrastructure using Proxmox with multiple Debian VMs and containers.
- Set up and maintains multiple services using Docker for personal use; began migration to a K8s cluster with Helm and declarative YAML configurations.
- Secured self-hosted services with TLS certificates, runs VPNs (WireGuard, ZeroTier), and Authentik;

Portfolio Website (erdaifuu.com) | *Next.js, React, Tailwind*

August 2025 - Present

- Developed a full-stack portfolio website using Next.js 14, TypeScript, and Tailwind CSS with Docker containerization for deployment
- Built custom UI component system with real-time position tracking and event handling for interactive windows.

TECHNICAL SKILLS

Languages: Python, Java, Ruby, Rust, Kotlin, HTML/CSS, TypeScript

Tools: AWS, Ruby on Rails, Docker, Kubernetes, Git, Linux, Proxmox, Firebase, Next.js, React, Tailwind

Certifications: AWS Certified Cloud Practitioner (2025)